

Advanced Image Processing: Assignment 4 (Due Apr 9, 2026)

Note: Please provide detailed comments for code that may be written to solve the following problems. The assignment will be evaluated not just based on the final results but also how you obtained them. Late submissions will be penalized.

Problem 1: 2-bit quantizer (20 points)

Consider a Laplacian source with pdf

$$f_X(x) = \frac{1}{6}e^{-\frac{|x|}{3}}$$

1. Find the optimal quantization points and decision boundaries for a 2-bit scalar uniform quantizer obtained by minimizing the mean squared error distortion. Also compute the distortion achieved by the optimal quantizer.
2. Find the quantization points and decision boundaries for a 2-bit Lloyd-Max quantizer obtained with any initialization of your choice. Compute the distortion achieved in this case.
3. Compare the distortion achieved by the uniform scalar quantizer and the Lloyd-Max quantizer and explain your observation.

Problem 2: Comparison of perceptual quality measures (25 points)

In this problem, you will compare PSNR and SSIM components with respect to their performance in terms of correlation with human perception.

A. Algorithms: Evaluate the following metrics on the dataset mentioned in Part (b). You should obtain a single number as quality with respect to each of the metrics below for each image.

1. Peak signal to noise ratio in pixel domain
2. SSIM version that only uses the luminance term
3. SSIM version that only uses the contrast term
4. SSIM version that only uses the structure term
5. Overall SSIM

B. Performance measurement: Download the database available at <http://ece.iisc.ac.in/~rajivs/courses/aip2016/hw5.rar>
The database comes with the following:

1. distorted images in the “gblur” folder
2. reference images in the “refimgs” folder
3. reference image name for every distorted image in the “gblur” folder in “refnames_blur”
4. human opinion scores in “blur_dmos”
5. indicator of whether the image in the “gblur” folder is an original image in “blur_orgs”

C. Questions

1. Compute the Spearman rank order correlation coefficient between the dmos scores in “blur_dmos” and each metric you considered in A after having removed the scores that correspond to the original images in the “gblur” folder (as mentioned earlier, this information is contained in blur_orgs).
2. Comment on the relative performances of all the indices

Problem 3: Object Detection and Real-World Robustness (25 points)

(a) Model Training: Train a YOLO-v8 model using the Vehicle Object Detection Dataset (5 classes). The corresponding annotation text files are provided in the YOLO format: `<class-id> <center-x> <center-y> <width> <height>`. Use the train, val and test sets provided. You can use a smaller subset of the data if the training takes too long.

(b) Evaluation: Evaluate the trained model on the held-out test set. Compute the following metrics: (i) mAP@0.5IoU; (ii) mAP@[0.5:0.95]IoU (take average of mAP at 10 IoU thresholds from 0.5 to 0.95 with a step size of 0.05). Also, provide few qualitative results (including both failure cases as well as where the model performs well) in your report.

(c) Domain Shift and Failure Analysis: Capture 10–15 original photographs of vehicles (cars, motorcycles, buses, etc.) around the campus using your smartphone. Preprocess the images similar to the test set images as before, and run inference on these “real-world” images using your trained model. Compare these results with the test set performance in Part (b). Explain why the trained model may fail or show lower confidence on these new test images. Consider factors such as *lighting conditions*, *viewpoint/angle variance*, *occlusion* when clicking the pictures.

Prepare a report containing the answers to all the problems and submit along with relevant code.